Happy Birthday to You — melody

Indexed concatenation, level by level

musIC · 05 October 2026

The piece

Happy Birthday to You — melody · 3/4 · 120 bpm · each set is {*pitch*, *duration*} · 25 sets

Each set is written {*pitch*, *duration*}:

pitch	MIDI note number: 60 = middle C, 0 = a rest
duration	how long the note lasts; 32 = a quarter note



The piece as a set, a bar to a line:

```
{{67,16},{67,16},{69,32},{67,32},
 {72,32},{71,64},
 {67,16},{67,16},{69,32},{67,32},
 {74,32},{72,64},
 {67,16},{67,16},{79,32},{76,32},
 {72,32},{71,32},{69,32},
 {77,16},{77,16},{76,32},{72,32},
 {74,32},{72,64}}
```

How far each level got

Level	works on	written as	
1	25 sets	12 parts	✓
2	12 items	3 parts	✓
3	3 items	—	0 of 3

Level 1: 25 sets → 12 parts

sets are written $\{pitch, duration\}$



In[1] sets 1–2 ✓ matches

$\in^2[\{67, 16\}]$

```
iC[{67, 16}, 2]
```

expands to 2 rows · written as 1 part

In[2] sets 3–4 ✓ matches

$\in_{i=0}^{1}[\{69 - 2i, 32\}]$

```
iC[{69 - 2*i, 32}, {i, 0, 1}]
```

expands to 2 rows · written as 1 part

In[3] sets 5–6 ✓ matches

$\in_{i=0}^{1}[\{72 - i, 32i + 32\}]$

```
iC[{72 - i, 32*i + 32}, {i, 0, 1}]
```

expands to 2 rows · written as 1 part

In[4] sets 7–8 ✓ matches

$\in^2[\{67, 16\}]$

```
iC[{67, 16}, 2]
```

expands to 2 rows · written as 1 part

In[5] sets 9–10 ✓ matches

$$\text{€}_{i=0}^{1}\left[\{69-2i, 32\}\right]$$

```
iC[{69 - 2*i, 32}, {i, 0, 1}]
```

expands to 2 rows · written as 1 part

In[6] sets 11–12 ✓ matches

$$\text{€}_{i=0}^{1}\left[\{74-2i, 32i+32\}\right]$$

```
iC[{74 - 2*i, 32*i + 32}, {i, 0, 1}]
```

expands to 2 rows · written as 1 part

In[7] sets 13–14 ✓ matches

$$\text{€}^{2}\left[\{67, 16\}\right]$$

```
iC[{67, 16}, 2]
```

expands to 2 rows · written as 1 part

In[8] sets 15–16 ✓ matches

$$\text{€}_{i=0}^{1}\left[\{79-3i, 32\}\right]$$

```
iC[{79 - 3*i, 32}, {i, 0, 1}]
```

expands to 2 rows · written as 1 part

In[9] sets 17–19 ✓ matches

$$\text{€}_{i=0}^{2}\left[\{-\frac{i\,(i+1)}{2}+72, 32\}\right]$$

```
iC[{-1/2*i*(i + 1) + 72, 32}, {i, 0, 2}]
```

expands to 3 rows · written as 1 part

In[10] sets 20–21 ✓ matches

$€^2\left[\{77, 16\}\right]$

```
iC[{77, 16}, 2]
```

expands to 2 rows · written as 1 part

In[11] sets 22–23 ✓ matches

$\overset{1}{\underset{i=0}{€}}\left[\{76 - 4i, 32\}\right]$

```
iC[{76 - 4*i, 32}, {i, 0, 1}]
```

expands to 2 rows · written as 1 part

In[12] sets 24–25 ✓ matches

$\overset{1}{\underset{i=0}{€}}\left[\{74 - 2i, 32i + 32\}\right]$

```
iC[{74 - 2*i, 32*i + 32}, {i, 0, 1}]
```

expands to 2 rows · written as 1 part

Level 1, all together ✓ matches

In[1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12] joined in the order of the sets they stand for

$\{€^2\left[\{67, 16\}\right], \overset{1}{\underset{i=0}{€}}\left[\{69 - 2i, 32\}\right], \overset{1}{\underset{i=0}{€}}\left[\{72 - i, 32i + 32\}\right], €^2\left[\{67, 16\}\right], \overset{1}{\underset{i=0}{€}}\left[\{69 - 2i, 32\}\right],$

$\overset{1}{\underset{i=0}{€}}\left[\{74 - 2i, 32i + 32\}\right], €^2\left[\{67, 16\}\right], \overset{1}{\underset{i=0}{€}}\left[\{79 - 3i, 32\}\right], \overset{2}{\underset{i=0}{€}}\left[\{-\frac{i\,(i+1)}{2} + 72, 32\}\right], €^2\left[\{77, 16\}\right],$

$\overset{1}{\underset{i=0}{€}}\left[\{76 - 4i, 32\}\right], \overset{1}{\underset{i=0}{€}}\left[\{74 - 2i, 32i + 32\}\right]\}$

```
{iC[{67, 16}, 2], iC[{69 - 2*i, 32}, {i, 0, 1}], iC[{72 - i, 32*i + 32}, {i, 0, 1}], iC[{67,
    16}, 2], iC[{69 - 2*i, 32}, {i, 0, 1}], iC[{74 - 2*i, 32*i + 32}, {i, 0, 1}], iC[{67, 16},
    2], iC[{79 - 3*i, 32}, {i, 0, 1}], iC[{-1/2*i*(i + 1) + 72, 32}, {i, 0, 2}], iC[{77, 16},
    2], iC[{76 - 4*i, 32}, {i, 0, 1}], iC[{74 - 2*i, 32*i + 32}, {i, 0, 1}]}
```

sets 1–25 · expands to 25 rows

Level 2: 12 items → 3 parts

sets are written $\{pitch, duration\}$



This level concatenates the 12 parts level 1 was written as:

```
IC2[{67,16}], IC(i,0,1)[{69-2i,32}], IC(i,0,1)[{72-i,32+32i}], IC2[{67,16}], IC(i,0,1)[{69-2i
    ,32}], IC(i,0,1)[{74-2i,32+32i}], IC2[{67,16}], IC(i,0,1)[{79-3i,32}], IC(i,0,2)[{72-i(i+1)
    /2,32}], IC2[{77,16}], IC(i,0,1)[{76-4i,32}], IC(i,0,1)[{74-2i,32+32i}]
```

In[1] items 1–6 ✓ matches

$$\overset{1}{\underset{j=0}{\in}}\left[\in^{2}\left[\{67,16\}\right],\overset{1}{\underset{i=0}{\in}}\left[\{69-2i,32\}\right],\overset{1}{\underset{i=0}{\in}}\left[\{-ij-i+2j+72,32i+32\}\right]\right]$$

```
iC[iC[{67, 16}, 2], iC[{69 - 2*i, 32}, {i, 0, 1}], iC[{-i*j - i + 2*j + 72, 32*i + 32}, {i, 0,
    1}], {j, 0, 1}]
```

expands to 12 rows · written as 1 part

In[2] items 7–11 ✓ matches

$$\overset{1}{\underset{j=0}{\in}}\left[\in^{2}\left[\{10j+67,16\}\right],\overset{1}{\underset{i=0}{\in}}\left[\{-ij-3i-3j+79,32\}\right],\overset{2-3j}{\underset{i=0}{\in}}\left[\{-\frac{i\,(i+1)}{2}+72,32\}\right]\right]$$

```
iC[iC[{10*j + 67, 16}, 2], iC[{-i*j - 3*i - 3*j + 79, 32}, {i, 0, 1}], iC[{-1/2*i*(i + 1) + 72,
    32}, {i, 0, 2 - 3*j}], {j, 0, 1}]
```

expands to 11 rows · written as 1 part

In[3] items 12 ✓ matches

$$\{\overset{1}{\underset{i=0}{\in}}\left[\{74-2i,32i+32\}\right]\}$$

```
{iC[{74 - 2*i, 32*i + 32}, {i, 0, 1}]}
```

expands to 2 rows · written as 1 part

Level 2, all together ✓ matches

In[1, 2, 3] joined in the order of the items they stand for

$$\{\overset{1}{\underset{j=0}{€}}\left[€^2\left[\{67,16\}\right]\right],\ \overset{1}{\underset{i=0}{€}}\left[\{69-2i,32\}\right],\ \overset{1}{\underset{i=0}{€}}\left[\{-ij-i+2j+72,32i+32\}\right]\right],\ \overset{1}{\underset{j=0}{€}}\left[€^2\left[\{10j+67,16\}\right],$$

$$\overset{1}{\underset{i=0}{€}}\left[\{-ij-3i-3j+79,32\}\right],\ \overset{2-3j}{\underset{i=0}{€}}\left[\{-\frac{i\,(i+1)}{2}+72,32\}\right]\right],\ \overset{1}{\underset{i=0}{€}}\left[\{74-2i,32i+32\}\right]\}$$

```
{iC[iC[{67, 16}, 2], iC[{69 - 2*i, 32}, {i, 0, 1}], iC[{-i*j - i + 2*j + 72, 32*i + 32}, {i, 0,
    1}], {j, 0, 1}], iC[iC[{10*j + 67, 16}, 2], iC[{-i*j - 3*i - 3*j + 79, 32}, {i, 0, 1}], iC
   [{-1/2*i*(i + 1) + 72, 32}, {i, 0, 2 - 3*j}], {j, 0, 1}], iC[{74 - 2*i, 32*i + 32}, {i, 0,
   1}]}
```

items 1–12 · expands to 25 rows

Level 3: 0 of 3 items written

sets are written $\{pitch, duration\}$



This level concatenates the 3 parts level 2 was written as:

```
IC(j,0,1)[IC(2)[{67,16}], IC(i,0,1)[{69-2i,32}], IC(i,0,1)[{72-i+2j-1(i*j),32+32i}]], IC(j,0,1)
   {IC(2)[{67+10j,16}], IC(i,0,1)[{79-3i-3j-1(i*j),32}], IC(i,0,2-3j)[{72-i(i+1)/2,32}]}, IC(i
   ,0,1)[{74-2i,32+32i}]
```

Result

Expanding level 2 gives back all 25 sets of the piece ✓.